

TRN006 – Cybersecurity Fundamentals

1. Course Overview

- **Duration:** 2 Days | 4 Hours per Day | Total 8 Hours
- **Delivery Mode:** Online Instructor-Led Training
- **Category:** Cybersecurity | Network Security | Compliance
- **Focus Areas:** Cybersecurity, OWASP, Network Security, Security Compliance
- **Target Audience:** IT Professionals, System Administrators, Developers, Network Engineers, Freshers, Security Enthusiasts

2. Course Objective

This program provides participants with foundational cybersecurity knowledge required to identify security threats, secure IT systems, understand network security principles, address web application vulnerabilities, and follow security best practices within enterprise environments.

3. Learning Outcomes

- Understand core cybersecurity principles and security terminology.
- Identify common cyber threats, attacks, and vulnerabilities.
- Apply security best practices for systems, networks, and applications.
- Understand OWASP Top 10 web application security risks.
- Implement essential security controls and compliance practices.
- Recognize phishing, malware, ransomware, and social engineering attacks.
- Improve security awareness and incident response readiness.

4. Curriculum Structure

DAY 1 – CYBERSECURITY FOUNDATIONS & NETWORK SECURITY

Session	Session 1: Introduction to Cybersecurity
Duration	45 Minutes
Topics Covered	Cybersecurity Basics, CIA Triad, Threat Landscape, Types of Cyber Attacks
Workshop / Activity	Cyber Threat Discussion
Expected Outcome	Understand cybersecurity fundamentals and modern threat environments.

Session	Session 2: Common Cyber Attacks
Duration	60 Minutes
Topics Covered	Phishing, Malware, Ransomware, Social Engineering, Insider Threats
Workshop / Activity	Attack Scenario Analysis
Expected Outcome	Recognize common attack vectors and risk indicators.

Session	Session 3: Network Security Fundamentals
Duration	60 Minutes
Topics Covered	Firewalls, VPN, IDS/IPS, Secure Network Architecture, Authentication
Workshop / Activity	Secure Network Design Workshop
Expected Outcome	Understand enterprise network protection techniques.

Session	Session 4: System & Endpoint Security
Duration	45 Minutes
Topics Covered	Password Policies, MFA, Endpoint Protection, Device Security
Workshop / Activity	Security Best Practices Exercise
Expected Outcome	Apply endpoint and account security controls.

Session	Session 5: Security Awareness & Best Practices
Duration	50 Minutes
Topics Covered	Safe Browsing, Email Security, Data Protection, Security Hygiene
Workshop / Activity	Security Awareness Quiz
Expected Outcome	Improve organizational cybersecurity awareness.

DAY 2 – APPLICATION SECURITY & COMPLIANCE

Session	Session 6: Web Application Security & OWASP
Duration	60 Minutes
Topics Covered	OWASP Top 10, SQL Injection, XSS, Broken Authentication, Secure Coding
Workshop / Activity	OWASP Vulnerability Review
Expected Outcome	Understand application security risks and mitigation techniques.

Session	Session 7: Security Monitoring & Incident Response
Duration	60 Minutes
Topics Covered	Security Logs, Incident Detection, Reporting, Response Lifecycle
Workshop / Activity	Incident Simulation Activity
Expected Outcome	Improve incident handling and response readiness.

Session	Session 8: Compliance & Governance
Duration	45 Minutes
Topics Covered	ISO 27001 Basics, GDPR Overview, Security Policies, Governance Frameworks
Workshop / Activity	Compliance Discussion
Expected Outcome	Understand security compliance and governance practices.

Session	Session 9: Cloud & Data Security Basics
Duration	45 Minutes
Topics Covered	Cloud Security Principles, Data Encryption, Access Management, Backup Strategies
Workshop / Activity	Cloud Security Exercise
Expected Outcome	Apply security concepts within cloud and enterprise systems.

Session	Session 10: Cybersecurity Capstone Simulation
Duration	50 Minutes
Topics Covered	Enterprise Security Scenario & Team Response
Workshop / Activity	Cybersecurity Team Simulation
Expected Outcome	Demonstrate practical cybersecurity analysis and response capabilities.

5. Hands-On Activities & Workshops

- Cyber Threat Identification Exercise
- Phishing Email Detection Activity
- Secure Network Design Workshop
- OWASP Vulnerability Analysis
- Security Awareness Quiz
- Incident Response Simulation
- Cloud Security Best Practices Exercise
- Cybersecurity Team Capstone Simulation

6. Assessment Framework

Assessment Component	Weightage
Hands-On Activities	30%
Knowledge Checks & Quizzes	20%
Security Workshops	20%
Final Cybersecurity Simulation	30%

7. Final Capstone Simulation

Participants work in teams to analyze and respond to a simulated cybersecurity incident affecting an enterprise environment.

- Identify security threats and affected systems.
- Analyze the root cause and vulnerabilities.
- Recommend security controls and mitigation actions.
- Prepare an incident response plan.
- Present findings and security recommendations.

8. Expected Deliverables

- Cybersecurity Risk Assessment Report
- Network Security Review Document
- OWASP Vulnerability Analysis Summary
- Incident Response Plan
- Cybersecurity Awareness Recommendations

9. Expected Business Outcomes

- Improve enterprise cybersecurity awareness.
- Reduce risks related to phishing and cyber attacks.
- Strengthen network and application security practices.
- Enhance incident response preparedness.
- Support compliance and governance initiatives.
- Build foundational cybersecurity capabilities across teams.

10. Training Methodology

Concepts & Security Theory: 40% | Hands-On Workshops & Simulations: 50% | Case Studies & Discussions: 10%